MIL-Q-45618F(MU)
26 March 1971
SUPERSEDING
MIL-Q-45618E(MU)
18 November 1963

MILITARY SPECIFICATION

QUADRANTS, FIRE CONTROL, ELEVATION: M13 SERIES

1. SCOPE

1.1 This specification covers elevation quadrants of the M13 Series. They *are* indirect fire control instruments used for setting gun elevation and depression angles. The model number indicates minor design variations in each quadrant.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a **part of this Specification to the extent specified herein.**

SPECIFICATIONS

Military

MIL-F-13926	**Fire Control Materiel, General Specification Governing the Manufacture and Inspection of**
MIL-P-14232	**Parts, Equipment and Tools for Army Materiel, Packaging and Packing of**
MIL-1-45607	**Inspection Equipment, Acquisition, Disposition and Maintenance of**

STANDARDS

Military

MIL-STD-105	**Sampling Procedures and Tables for Inspection by Attributes**
MIL-STD-109	**Quality Assurance Terms and Definitions**

DRAWINGS

U.S. Army, Frankford Arsenal

D6510977	Elevation Test Fixture **Assembly**
C7569504	Test Fixture, Accuracy
F7695046	Quadrant, Fire Control **Elevation: M13A3**
F8270843	Quadrant, Fire Control **Elevation: M13B1**

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F8286737	Quadrant, Fire Control Elevation: M13AlC
F8619627	Quadrant, Fire Control Elevation: M13A2
F8642402	Quadrant, Fire Control Elevation: M13Al

PACKAGING DATA SHEETS

7652218	Packaging of Quadrants, Fire Control, M13A1 and M13A1C
8619627	Packaging of Quadrants, Fire Control, M13A2, M13A3 and M13B1

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The quadrant shall be manufactured in accordance with the drawing applicable to the type quadrant specified in the contract and drawings pertinent thereto.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c* Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Operating temperatures.- The quadrant shall meet the requirements of 3.6.4 while exposed to and thermally stabilized at ambient temperatures of -40°F and +150°F. Subsequent to this exposure, the quadrant shall be thermally stabilized at standard ambient temperatures {+60° to +90°F), and shall meet the requirements of 3.4.2 and 3.6.

3.4.2 Storage temperatUre.- The quadrant shall show no evidence of physical failure such as casting failure or glass breakage and shall **meet** the applicable requirements of this specification at a standard ambient temperature of 60° to 90°F after having been exposed to and thermally stabilized at -80° and +160°F.

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3.4.3 Vibration.- The quadrant shall show no evidence of physical failure and shall meet the requirements of 3.6 after being subjected to a vibration of 30 cycles per second at an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds.

3.5 Alignment.- With the quadrant setup as shown on the respective drawing, the mounting pad surfaces shall be in a vertical plane, the key slots horizontal, the elevation and micrometer scales set at *zero,* and the level bubble central with respect to its graduations.

3.6 Performance.

3.6.1 Accuracy.- At any setting between 0 and -200 mils depression, and between 0 and +600 roils elevation, the error shall not exceed 1.0 roil, excluding backlash.

3.6.2 Backlash.- Backlash in the elevation mechanism shall not exceed 0.5 mil at any setting.

3.6.3 Level vial.- The level bubble shall remain central with respect to its graduations within one-half graduation (0.05-inch bubble movement) when the quadrant is rotated 15 degrees about its horizontal key slots, each side of its vertical position.

3.6.4 Torque.- Torque required to rotate the elevation micrometer knob shall not exceed 5 pound inches at all temperatures specified in 3.4.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any **other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are necessary to assure supplies and services conform to prescribed requirements.**

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of Inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contrac-*tor)* for first article testing shall be as specified in the contract.

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The sample for first article approval tests shall consist of 3 quadrants plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of Product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPS, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, 11 and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I, 11 and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR : AQL 0.65% Defective		
101. Vibration	3.4.3	4.6.3
102. Alignment	3.5	4.6.4
103. Accuracy	3.6.1	4.6.5
104. Backlash	3.6.2	4.6.6
105. Level vial	3.6.3	4.6.7
106. Torque	3.6.4	4.6.8
MINOR : NONE		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- One quadrant shall be selected at random from each 100 produced or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table II.

TABLE 11

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Materials	3.1	Drawings and specifications
302. Fabrication	3.2	Visual and drawings
303. General specifications	3.3	Visual and MIL-F-13926

4.4.1 Environmental.- Three quadrants shall be selected at random as a special sample from each 50 produced *or* from each month's production whichever occurs first. The samples shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304. Operating temperatures	3.4.1	4.6.2
305. Storage temperatures	3.4.2	4.6.1

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications-or SQAP forming a-part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection documents when provided with the technical data package.

4.5.2.2 contractor design.- The contractor shall design and supply inspection equipment comparable with the "Test Methods and Procedures" specified in 4.6 of this specification and with the applicable component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements Of MIL-F-13926 and the conditions of 3.4.2. The quadrant shall be exposed to each temperature extreme specified in 3.4.2 for a minimum of 5 hours. After each temperature exposure, the quadrant shall be stabilized in an ambient temperature between +60° and +90° Fahrenheit (F) and tested for conformity with the requirements of 3.6. Caution shall be exercised during this test to avoid subjecting the quadrant to thermal shock.

4.6.2 Extreme operating temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The quadrant shall be exposed to each temperature extreme specified in 3.4.1 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the quadrant shall be subjected to the applicable test of 4.6.8. Caution shall be exercised during this test to avoid subjecting the quadrant to thermal shock.

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4.6.3 vibration.- Standard vibrating equipment capable of producing the amplitude and frequency specified in 3.4.3 shall be utilized for this test. 'Vibration shall be in the vertical plane. Unless otherwise specified, the test equipment for vibration shall be in conformance with the "Test Facilities" requirement of MIL-F-13926.

4.6.3.1 post-vibration.- Subsequent to vibration, the quadrants shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4.3 prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts prior to being subjected to the remaining performance tests. Minimum visual examination shall include *the level* vial *(also* use smell test to detect leaking vial resulting from broken seal or glass), screws, nuts, etc. A tactile examination shall also be conducted to determine looseness or binding or moving parts such as worm gear, rotating knob, etc. Any one of these possible defects defined as such, shall be cause to consider a quadrant as defective. Quadrants which have passed the above examinations shall be tested for all the requirements of this specification.

4.6.4 Alignment.- The quadrant under test shall be properly positioned in Test Fixture c7569504 and Elevation Test Fixture Assembly D6510977 which shall be placed on a firm foundation and carefully leveled. The quadrant shall be aligned in accordance with and shall meet the requirements specified in 3.5.

4.6.5 Accuracy.- Set the quadrant to 400 roils elevation as indicated on the scale and micrometer, and then mount it on the test fixtures specified in 4.5.3. Rotate the mounting surface of Test Fixture D6510977 until the quadrant level bubble is central with respect to its graduations. The measured angle of test fixtures D6510977 mounting surface shall indicate the quadrant setting with the tolerance specified in 3.6.1. Elevation accuracy shall be measured in 100 mil increments from minus 200 to plus 600 mils. Measurements shall be made with backlash excluded for direction of travel.

4.6.6 Backlash.- This test may be conducted concurrent with the test specified in 4.6.5. The quadrant shall be mounted on the test fixtures specified in 4.6.4 and tested for backlash at 200 mil increments throughout the entire excursion of the quadrant. Backlash is the difference in a quadrant reading taken at each of these points, in an elevation direction when the bubble is central with respect to a graduation, and the same quadrant reading taken in a depression direction, when the level bubble is again central with respect to the graduation. The difference can be read on the test fixture, D6510977, and shall not exceed the tolerance specified in 3.6.3.

4.6.7 Level vial.- Set the quadrant to zero roils elevation as indicated on scale and micrometer, and then mount it on the test fixtures as specified **in 4.6.4. Rotate the mounting surface of the equipment until**

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the **quadrant level bubble is central with respect to graduations. The position of the bubble shall not change in excess of the allowable tolerance when the quadrant is canted as specified in 3.6.3.**

4.6.8 Torque.- A commercial type inch-pound torque wrench with an adapter to accommodate the elevation micrometer knob shall be used. Readings with the torque wrench shall be taken in both directions at the temperatures specified in 3.4.1. The torque required to rotate the elevation knob shall not exceed the value as specified in 3.6.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with the applicable packaging data sheet listed in Section 2. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing (5.1)
c. Applicable packaging data sheet number (see 5.1)
d. Applicable stock number

Custodian:

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Preparing activity:

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Project No. 1290-A285

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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